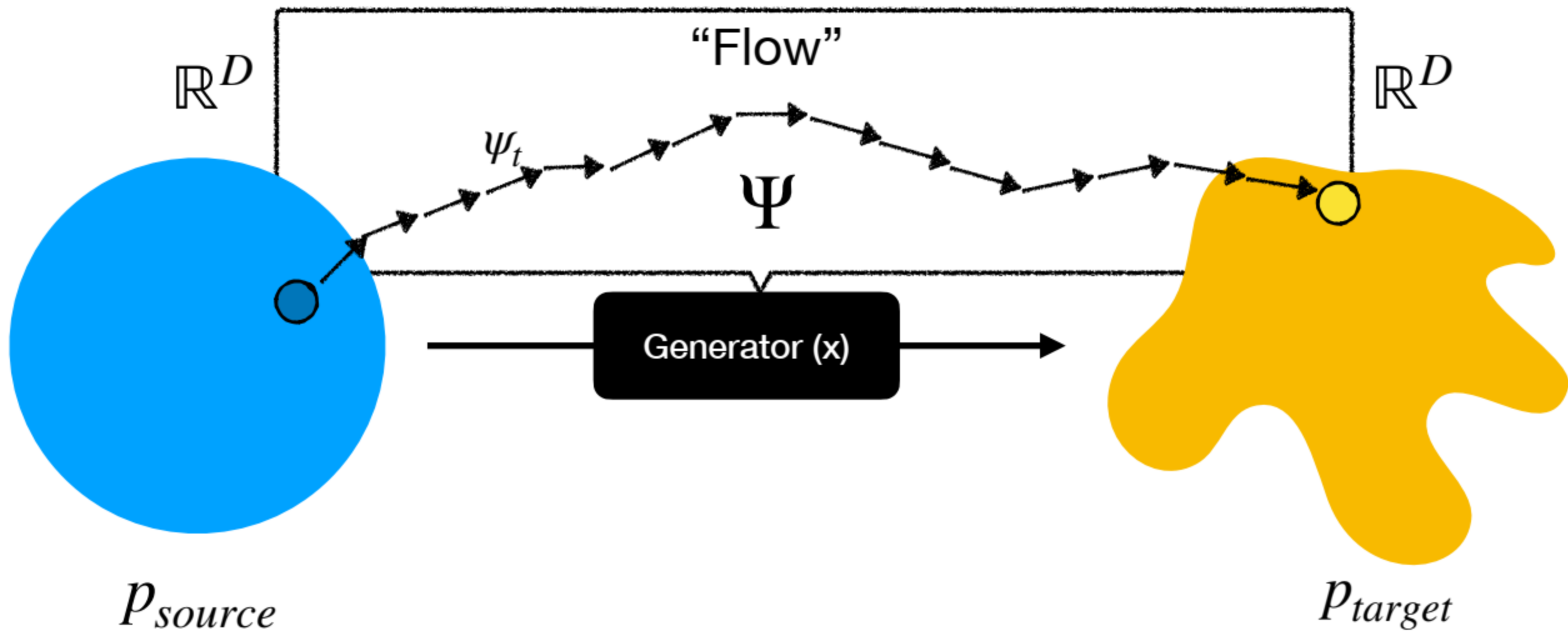


Flow-Matching as a Layer

Simon Korman

CVLAB Summer 2026

Generative Flow-Matching (FM)



FM training

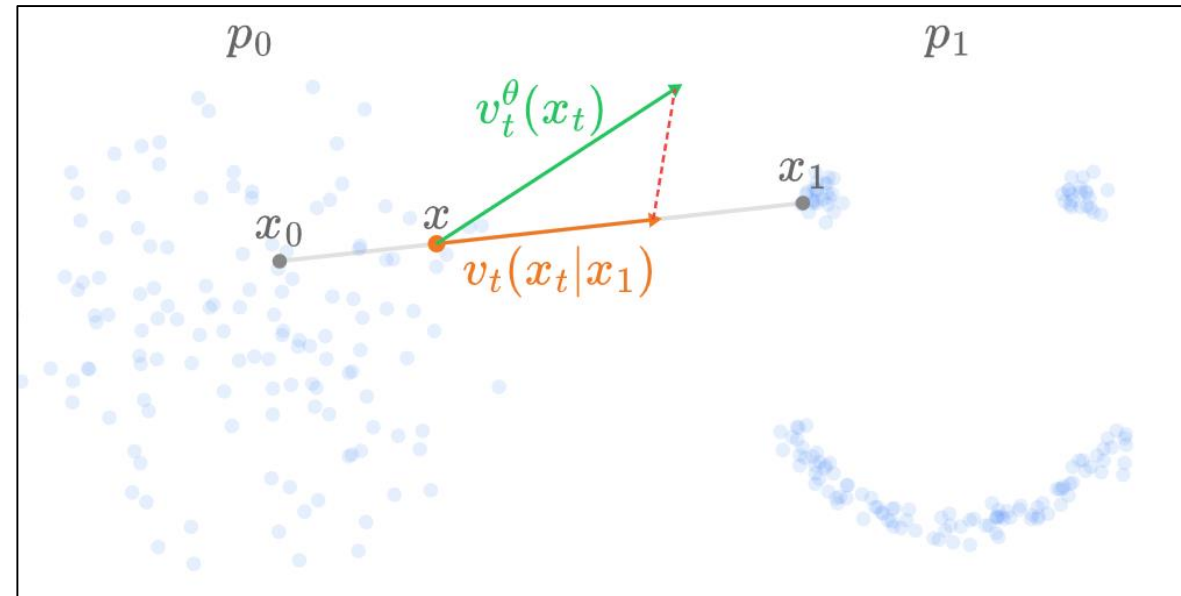
Algorithm 1: Unconditional training

Input: data p_0 , VAE encoder $\mathcal{E}$, velocity estimator v_θ , learning rate η ;

repeat

$\mathbf{x}_0 \sim p_0, \mathbf{z}_0 \leftarrow \mathcal{E}(\mathbf{x}_0)$;
 $\mathbf{z}_1 \leftarrow \mathcal{N}(0, \mathbf{I}), t \sim \mathcal{U}[0, 1]$;
 $\mathbf{z}_t \leftarrow (1 - t)\mathbf{z}_0 + t\mathbf{z}_1$;
 $l \leftarrow \|\mathbf{z}_1 - \mathbf{z}_0 - v_\theta(\mathbf{z}_t, t)\|_2^2$;
 $\theta \leftarrow \theta - \eta \nabla_\theta l$;

until convergence;



FM inference

Algorithm 2: Unconditional Euler sampling

Input: *initial noise* $\mathbf{z}_1$, *VAE decoder* $\mathcal{D}$,
velocity estimator v_θ , *steps* N ;

$t_n \leftarrow \frac{n}{N}$;

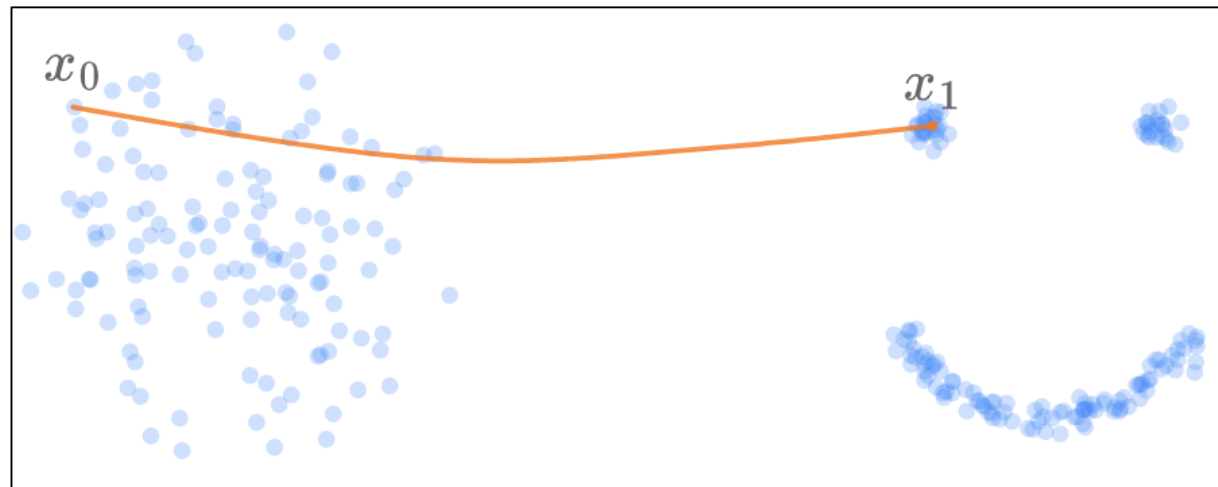
for $n = N - 1$ **to** 0 **do**

$\mathbf{z}_{t_n} \leftarrow \mathbf{z}_{t_{n+1}} + (t_n - t_{n+1})v_\theta(\mathbf{z}_{t_{n+1}}, t_{n+1})$

end

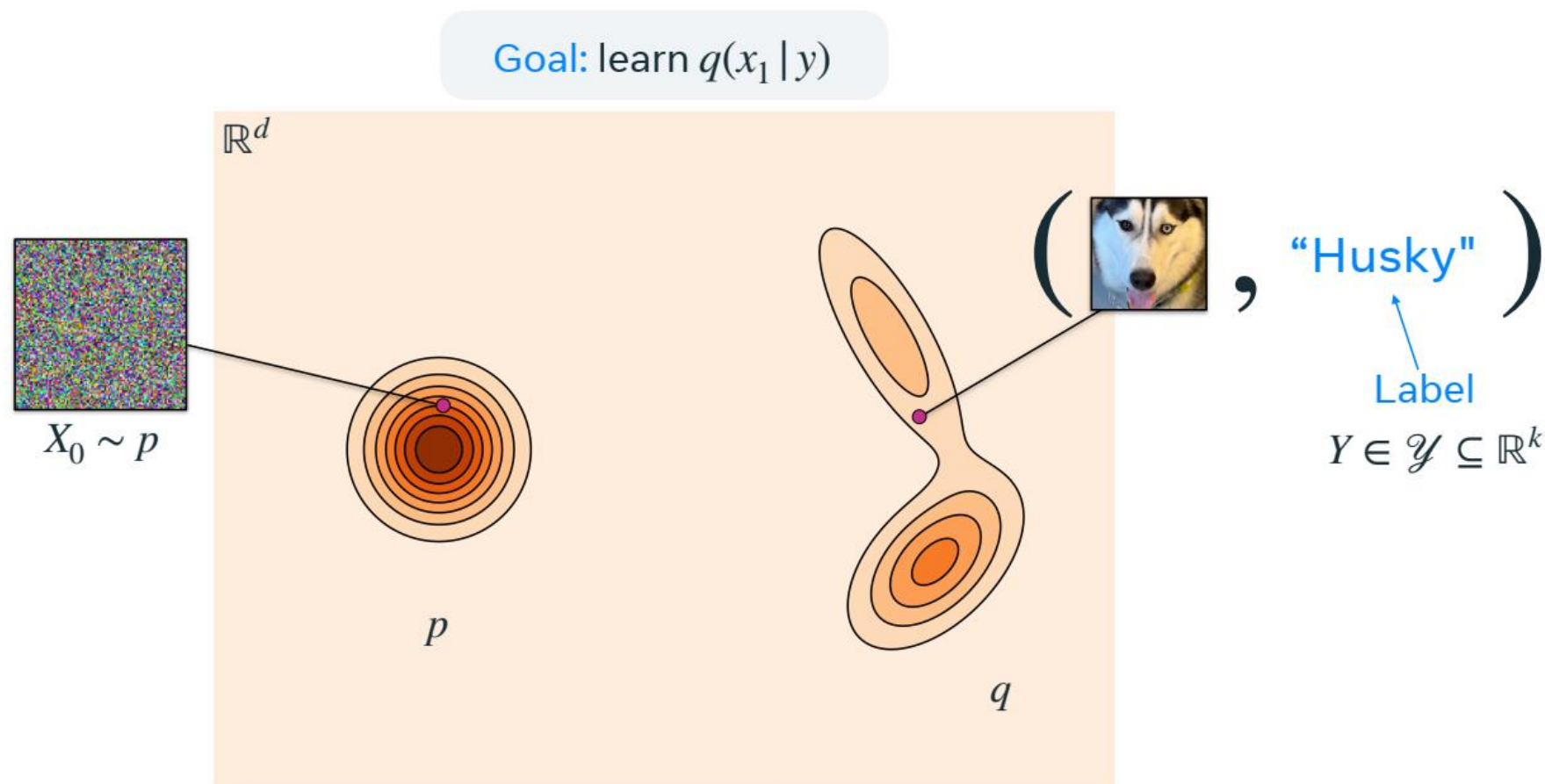
$\mathbf{x}_0 \leftarrow \mathcal{D}(\mathbf{z}_0)$;

Output: $\mathbf{x}_0$;



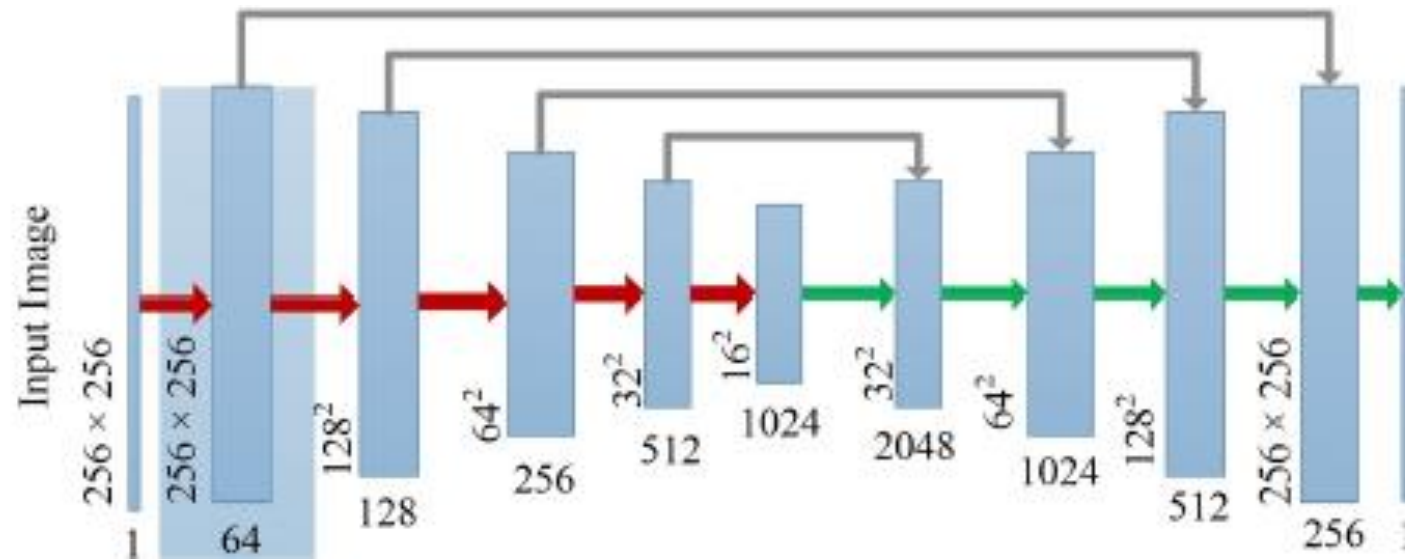
FM for generative tasks

- Noise to data
- Unpaired
- (Un)conditional



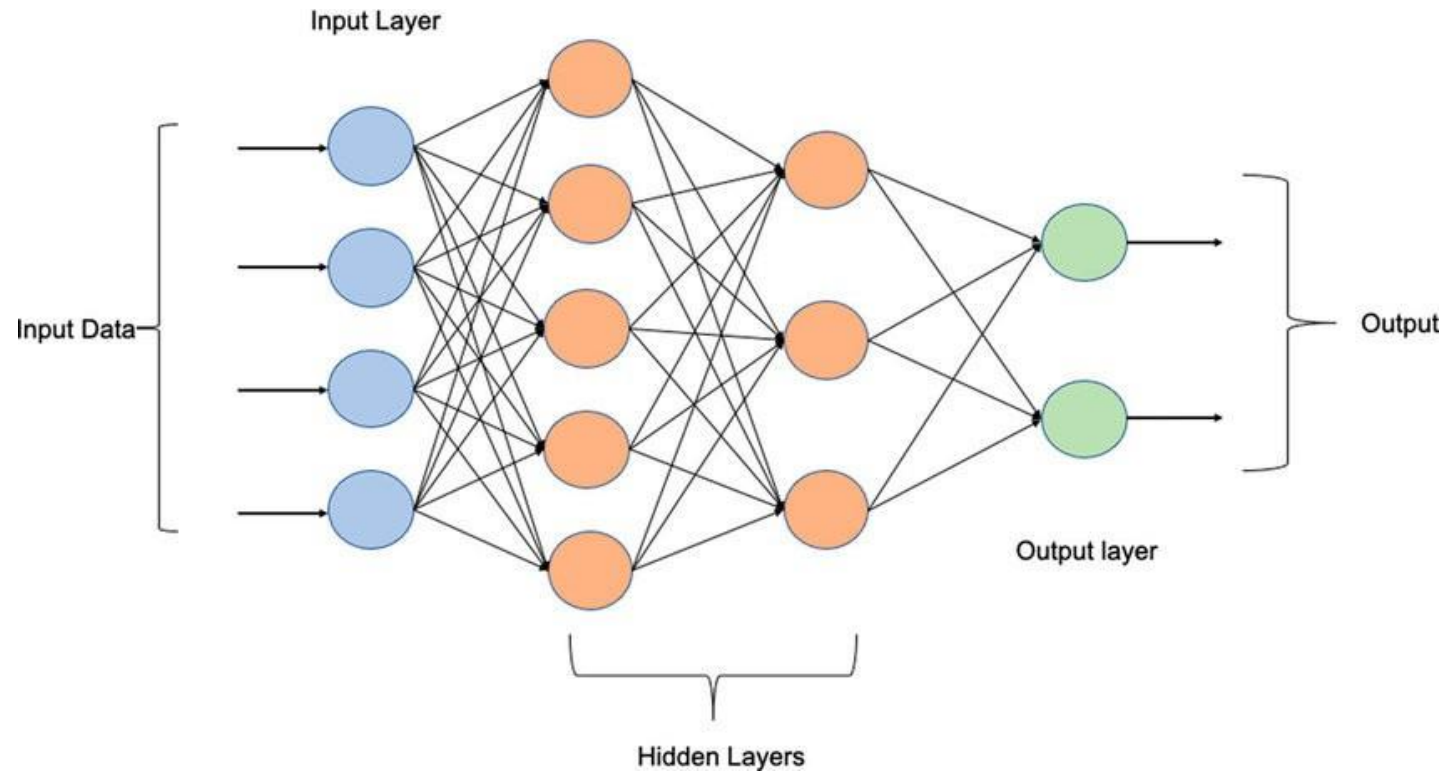
FM Network Architectures

image space



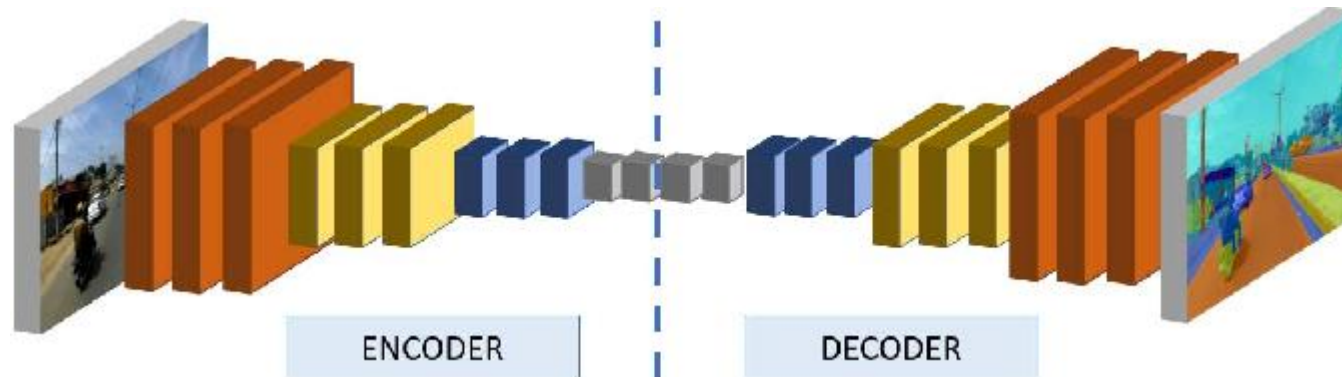
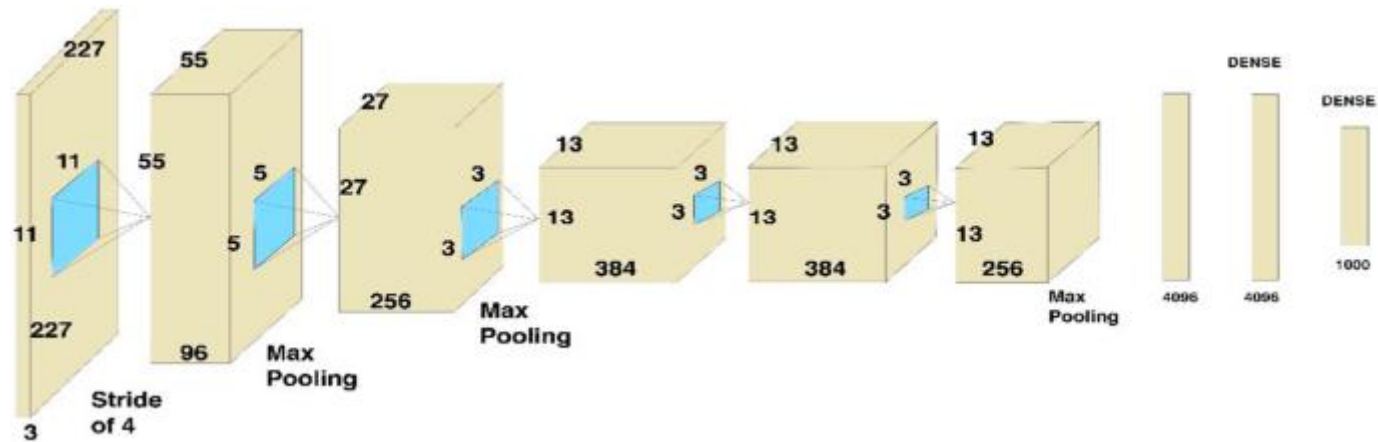
FM for generative tasks

latent space



FM for discriminative tasks

- Tasks: classification, segmentation, etc'

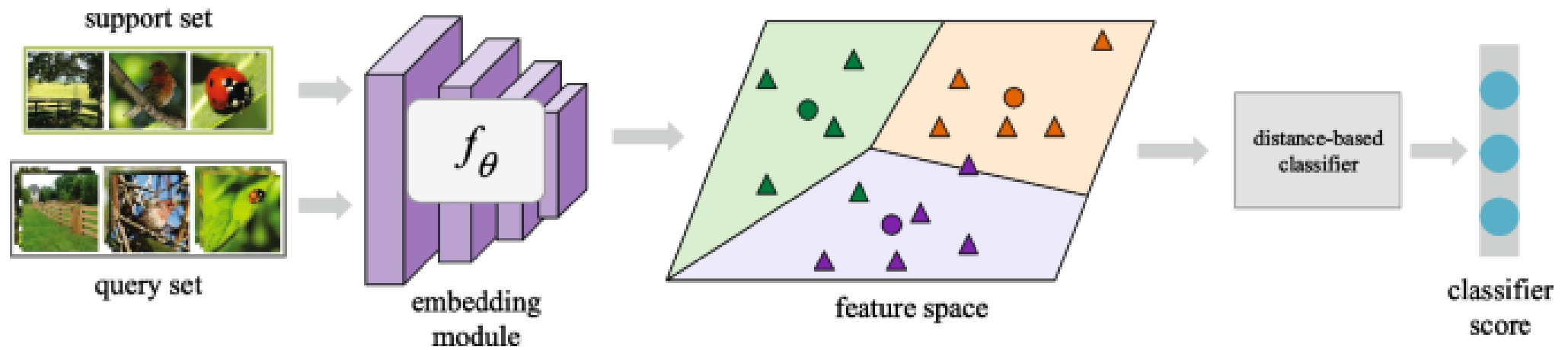


FM as a layer for discriminative tasks

- Replace standard network layers with FM
- Typically paired
- How to train?

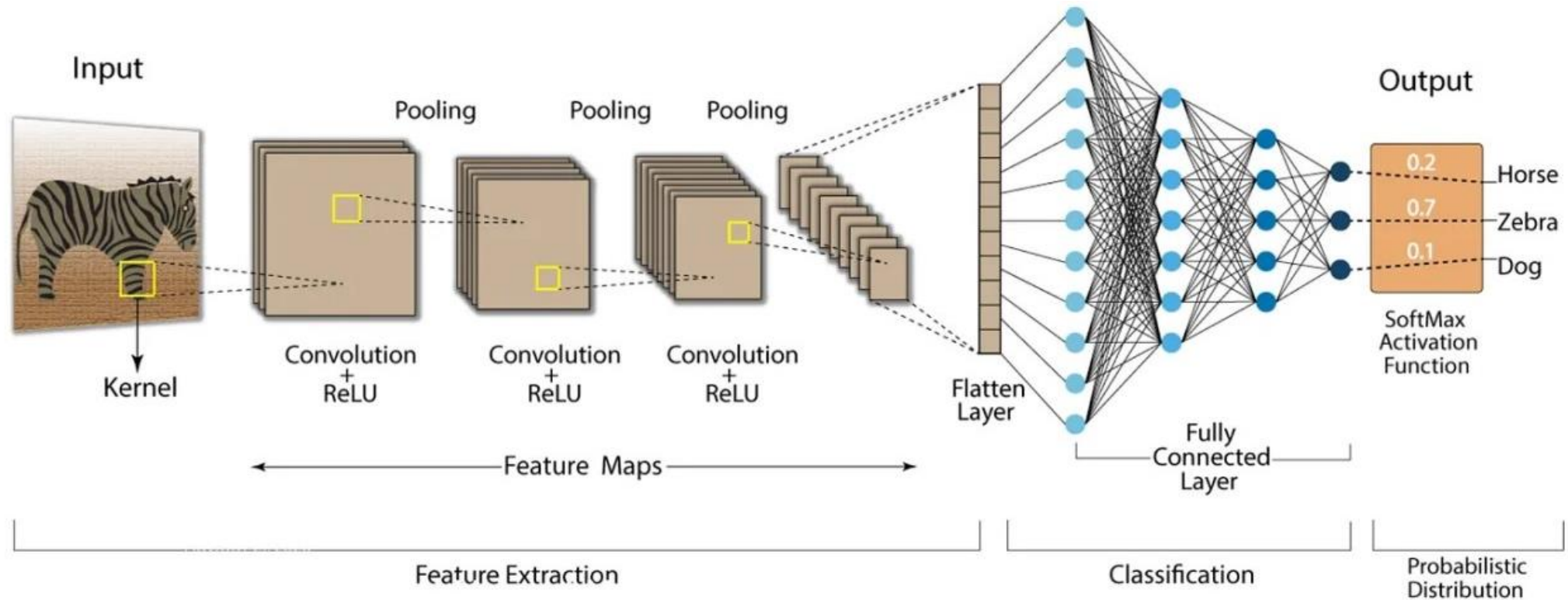
FM in a classifier

- 1. Distance to prototype based



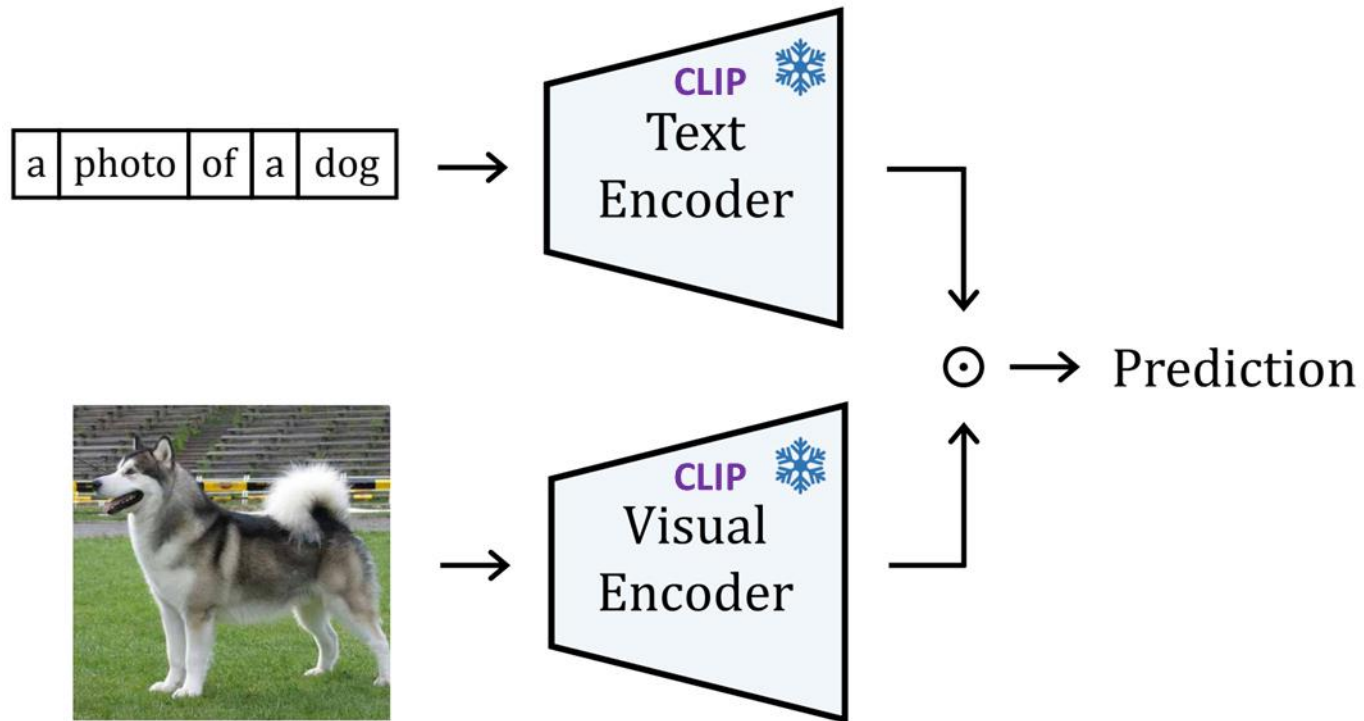
FM in a classifier

- 2. Linear Classifier head



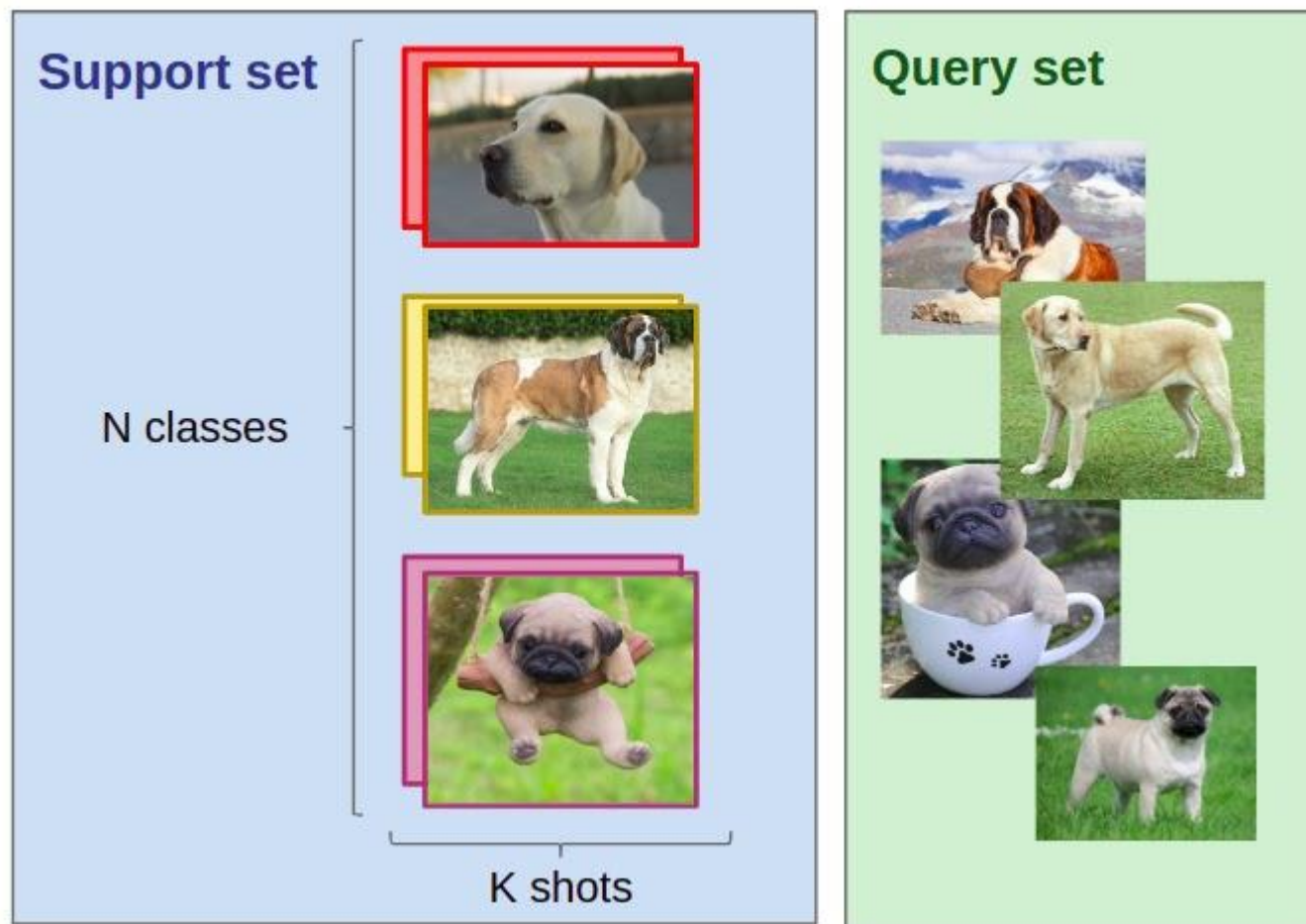
FM in a classifier

- 3. Zero-Shot CLIP



FM vs. iterative inference

Few-Shot Classification



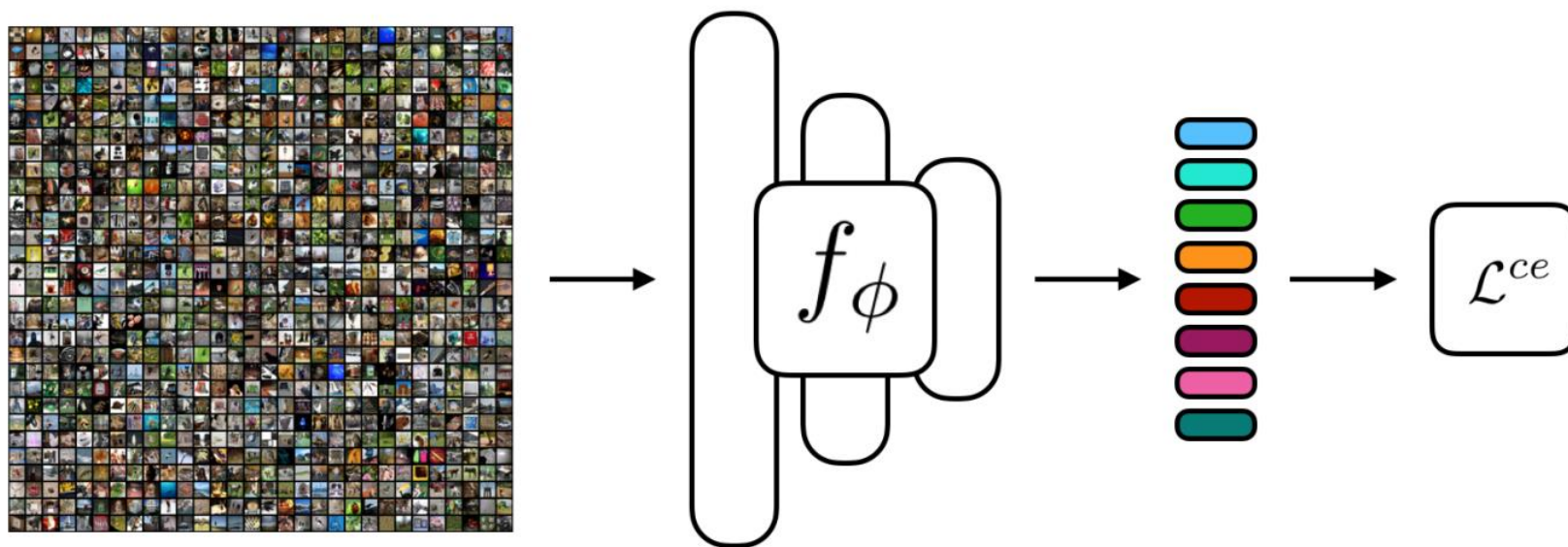
"Matching networks for one shot learning." (Vinyals et al. *NeurIPS* 2016).

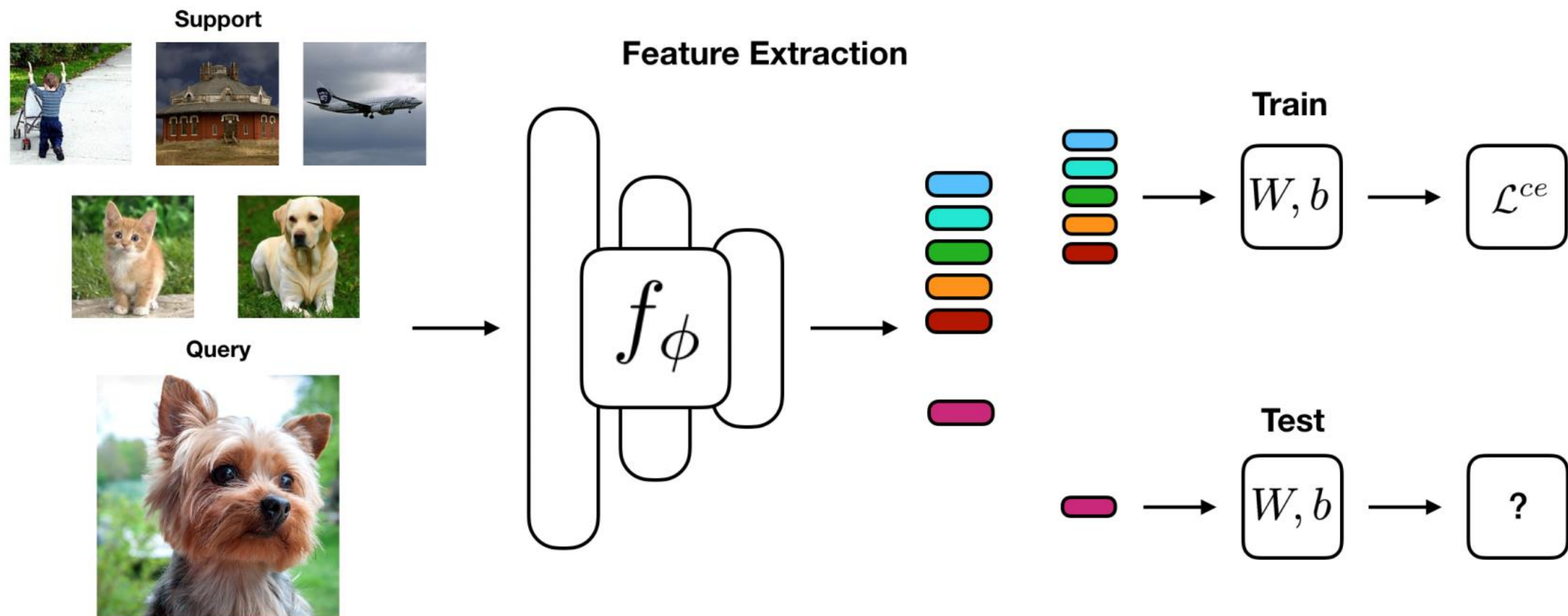
Rethinking Few-shot Image Classification: A Good Embedding is All You Need?

Yonglong Tian^{1*}, Yue Wang^{1*}, Dilip Krishnan², Joshua B. Tenenbaum¹, and Phillip Isola¹

¹ MIT

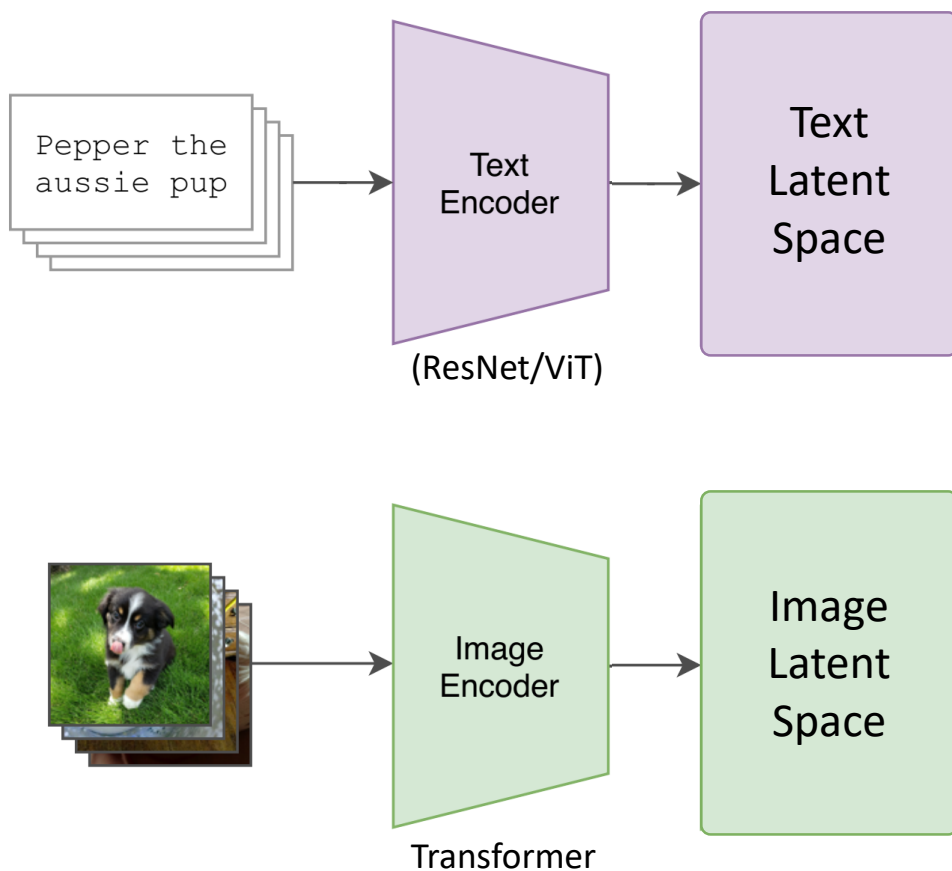
² Google Research





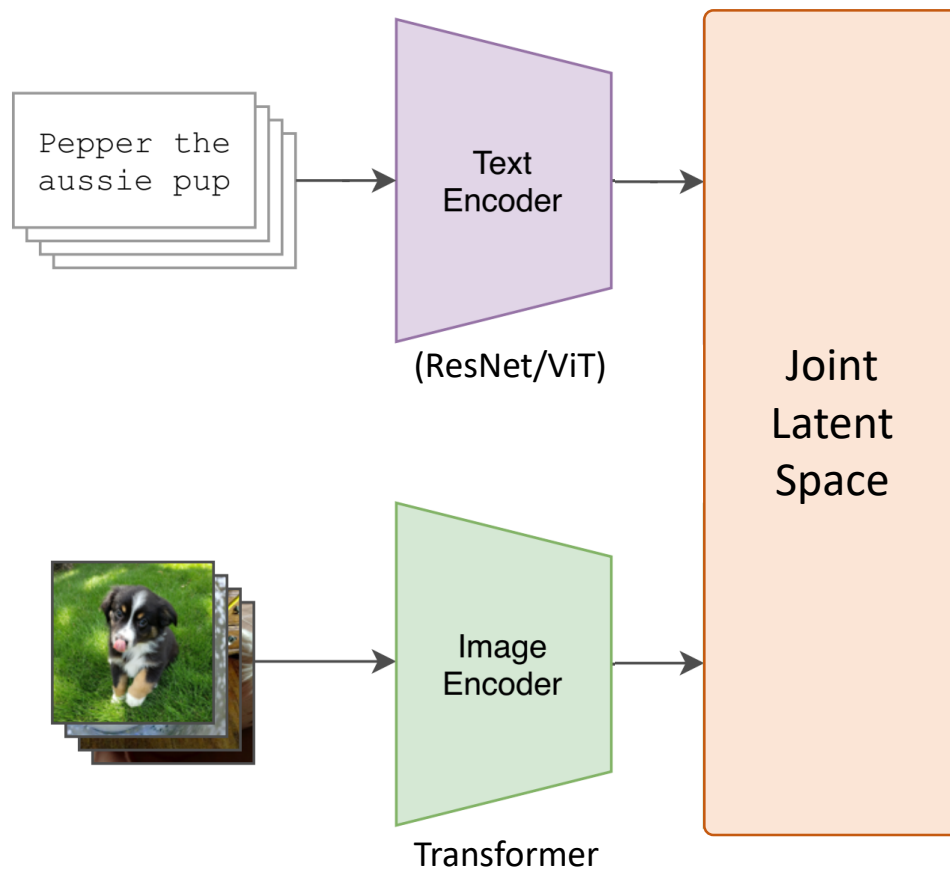
Learning Transferable Visual Models From Natural Language Supervision

Alec Radford^{*1} Jong Wook Kim^{*1} Chris Hallacy¹ Aditya Ramesh¹ Gabriel Goh¹ Sandhini Agarwal¹
Girish Sastry¹ Amanda Askell¹ Pamela Mishkin¹ Jack Clark¹ Gretchen Krueger¹ Ilya Sutskever¹



Learning Transferable Visual Models From Natural Language Supervision

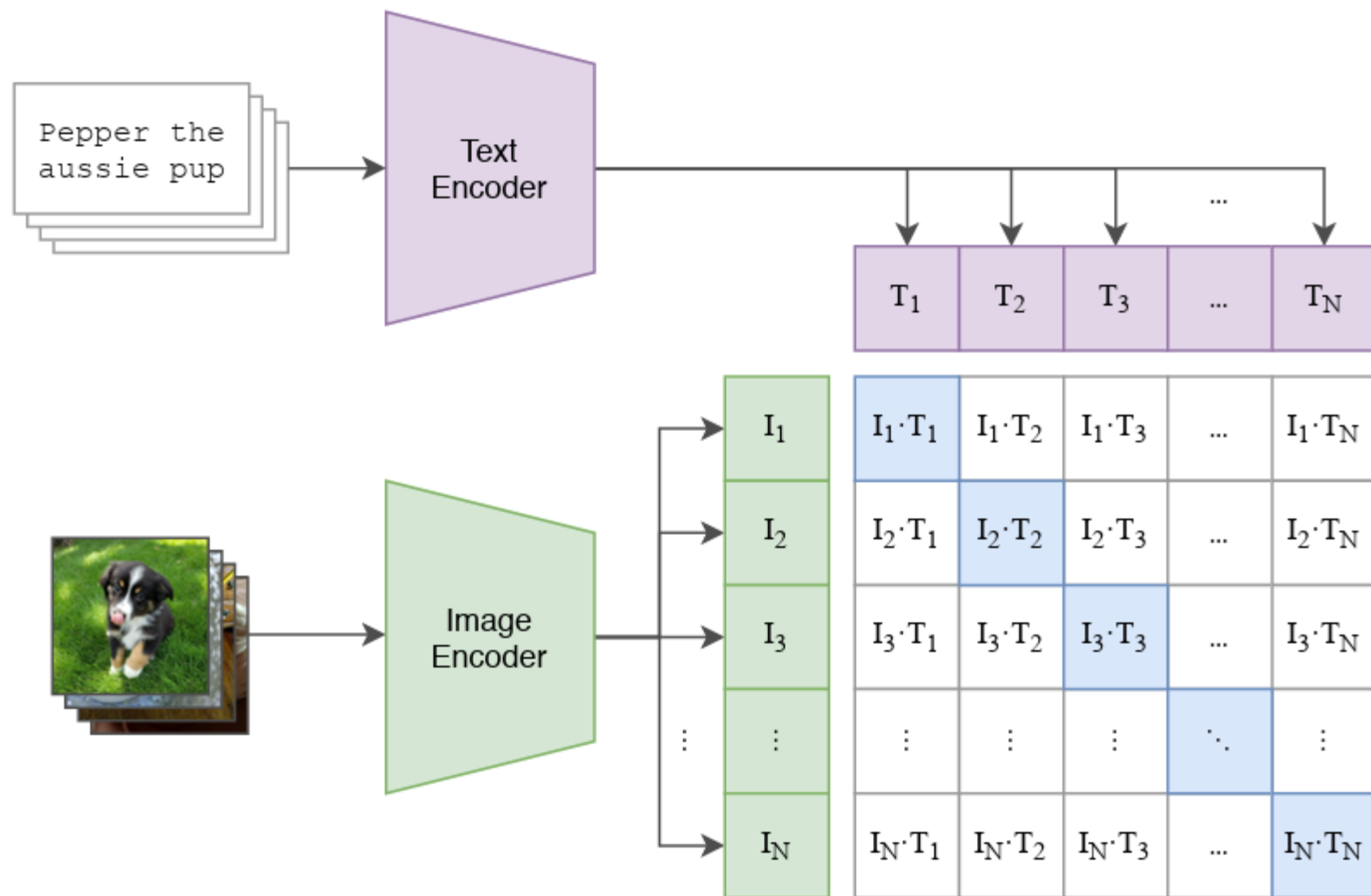
Alec Radford^{*1} Jong Wook Kim^{*1} Chris Hallacy¹ Aditya Ramesh¹ Gabriel Goh¹ Sandhini Agarwal¹
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A (joint) **multi-modal** encoder!

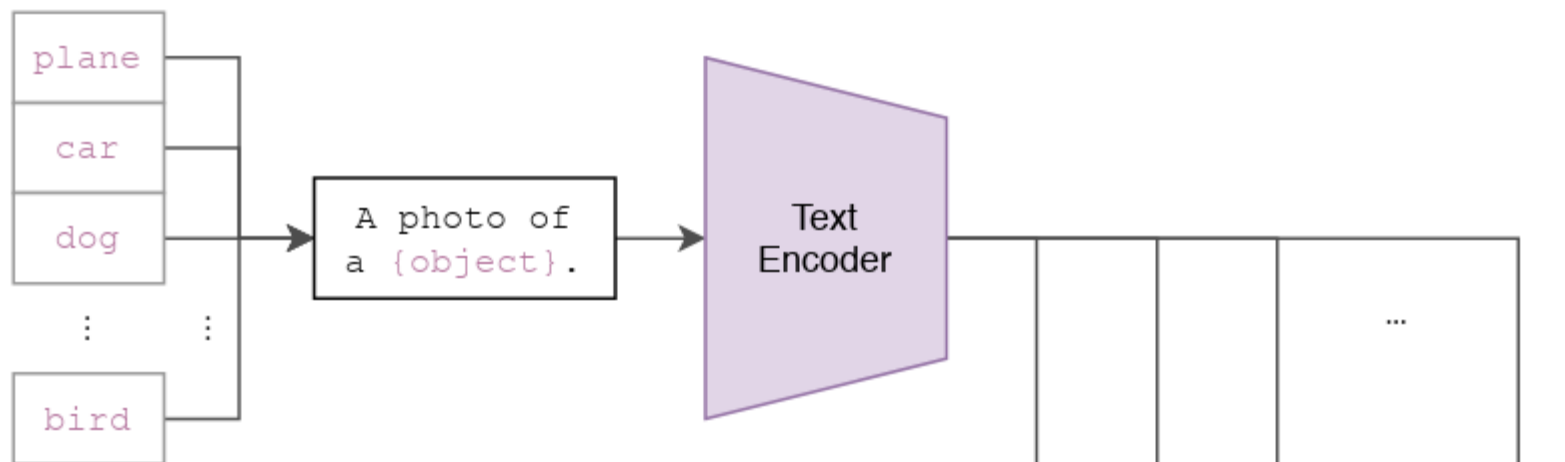
Zero-Shot CLIP

(1) Contrastive pre-training

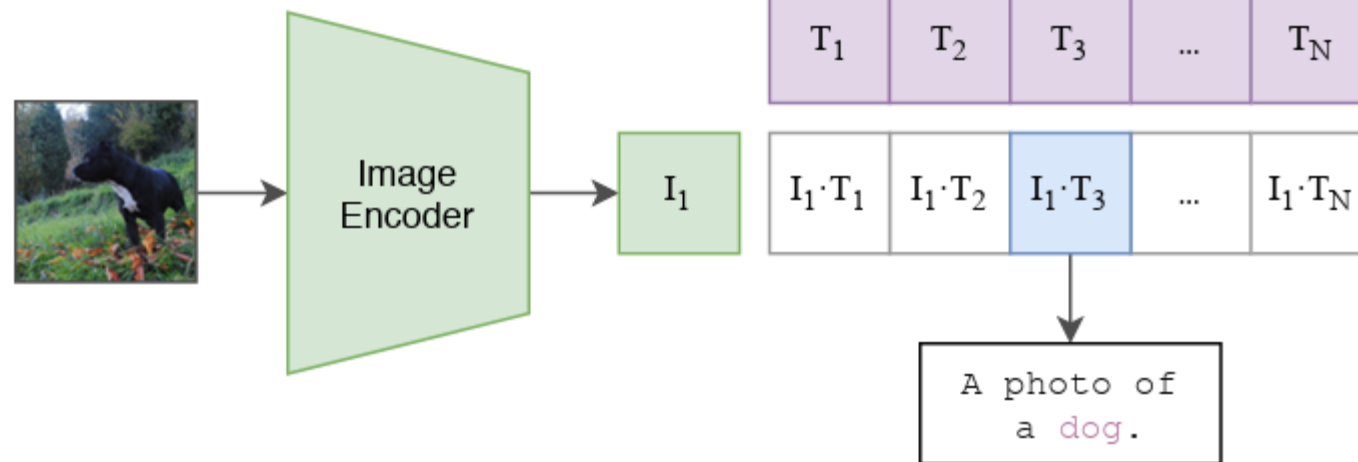


Zero-Shot CLIP

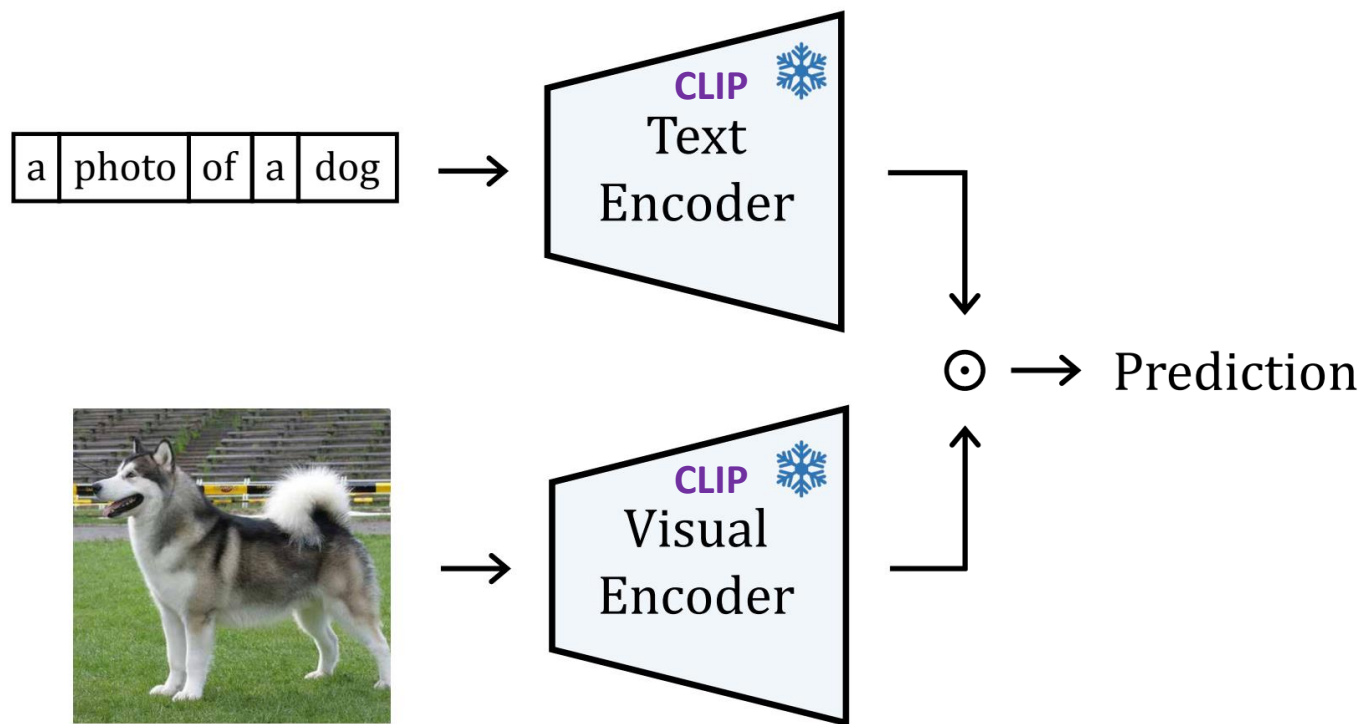
(2) Create dataset classifier from label text

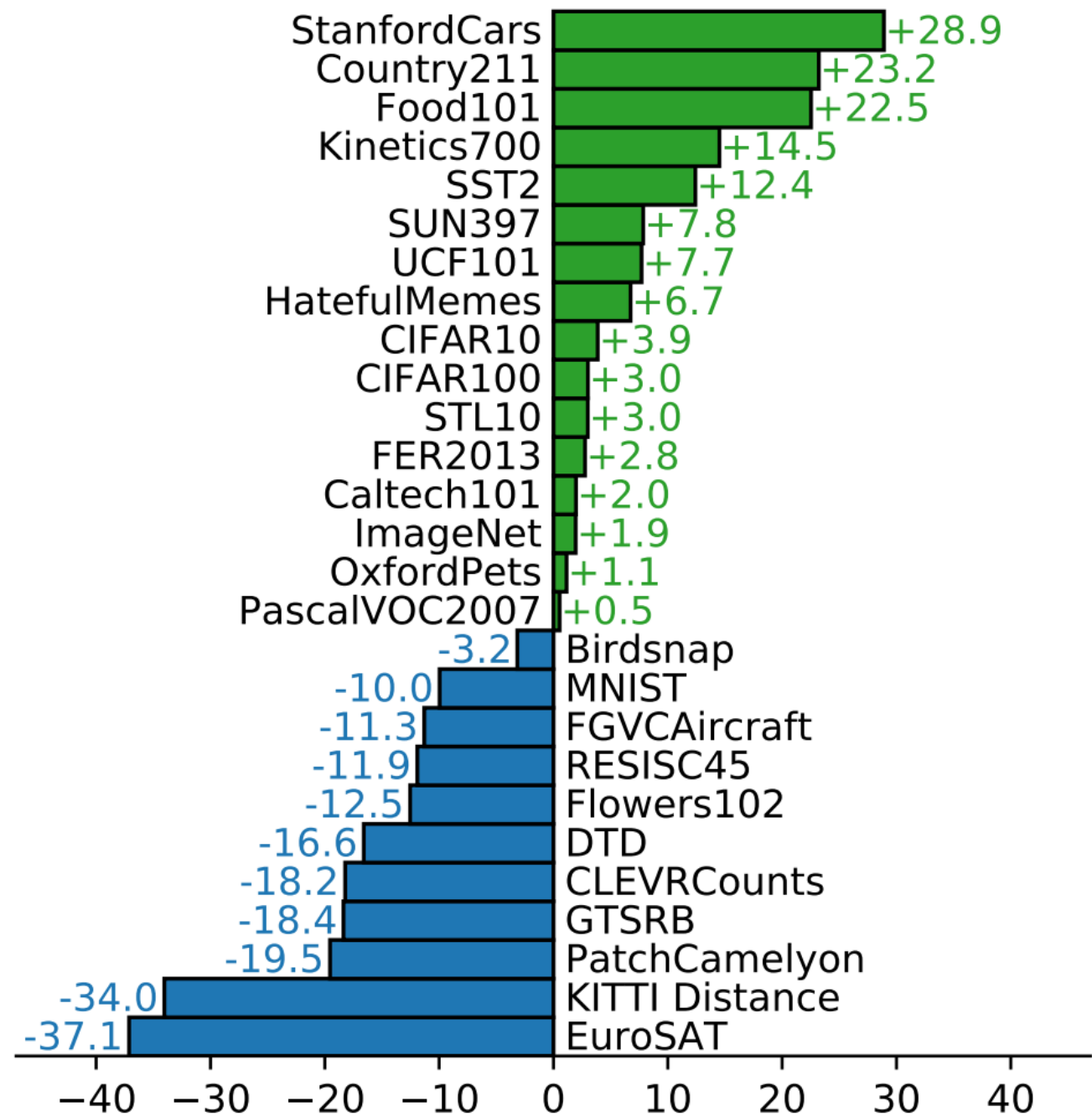


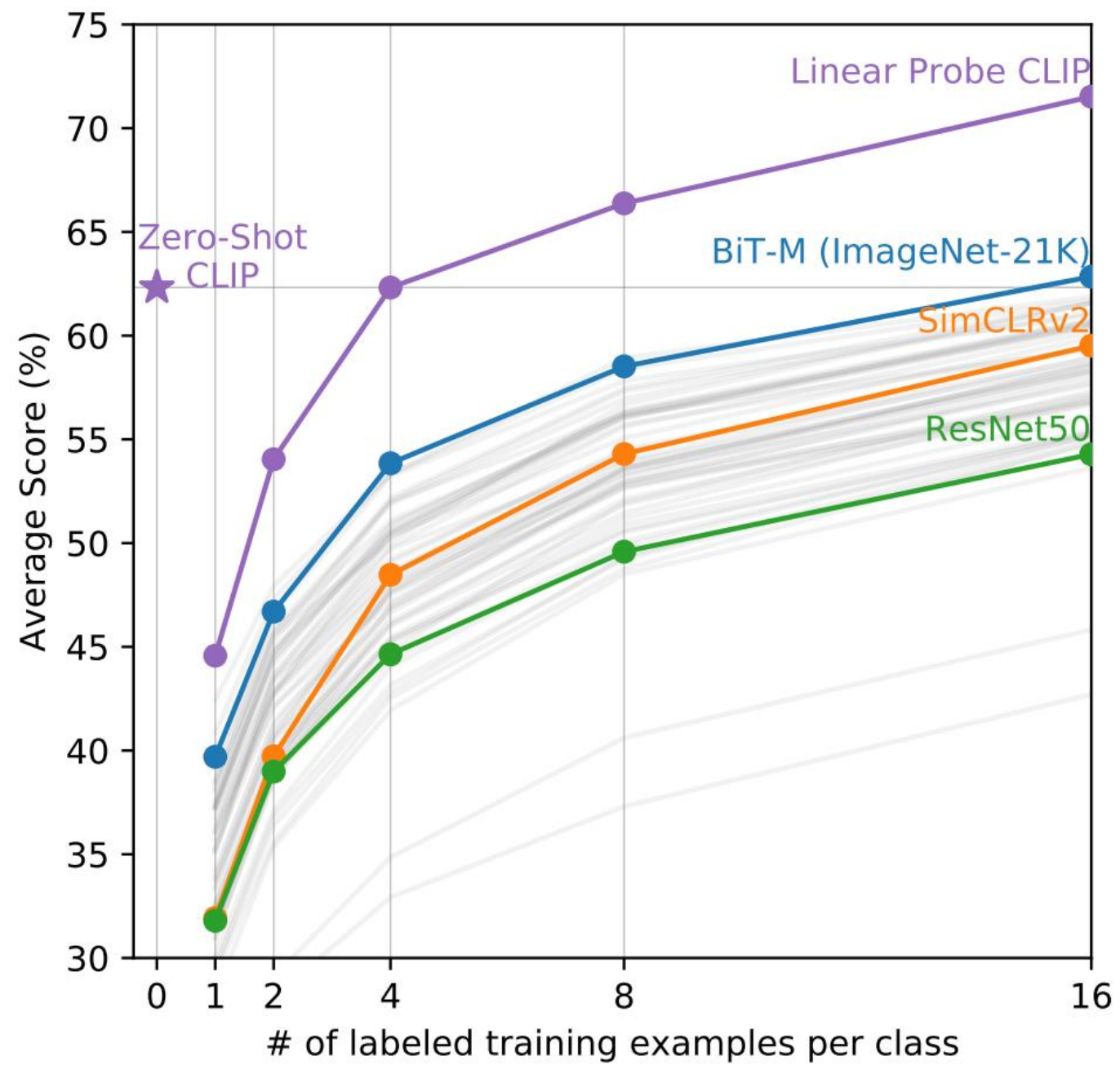
(3) Use for zero-shot prediction



Zero-Shot CLIP





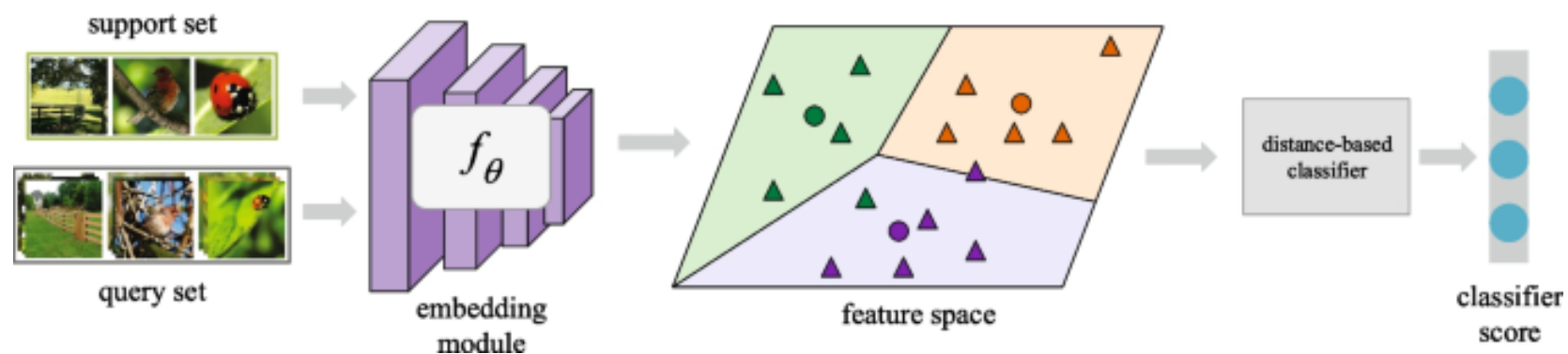


Stage 1: Baselines

- Few-Shot Classification
- MNIST, CIFAR, Mini-Imagenet
- K-means Prototype based
- Linear Classifier
- Zero-shot CLIP

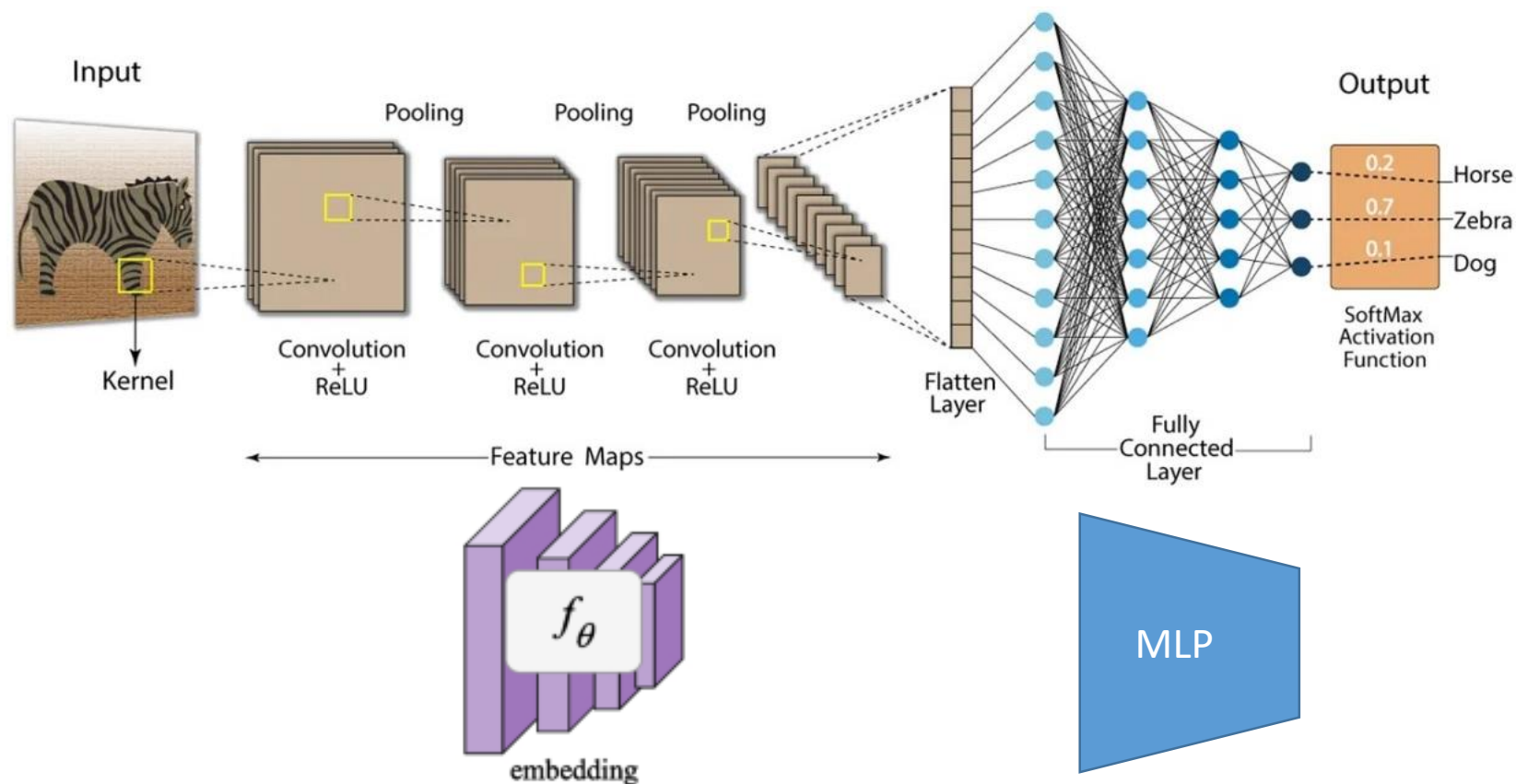
Stage 1: Baselines

- Few-Shot Classification
- MNIST, CIFAR, Mini-Imagenet
- **K-means Prototype based**
- Linear Classifier
- Zero-shot CLIP



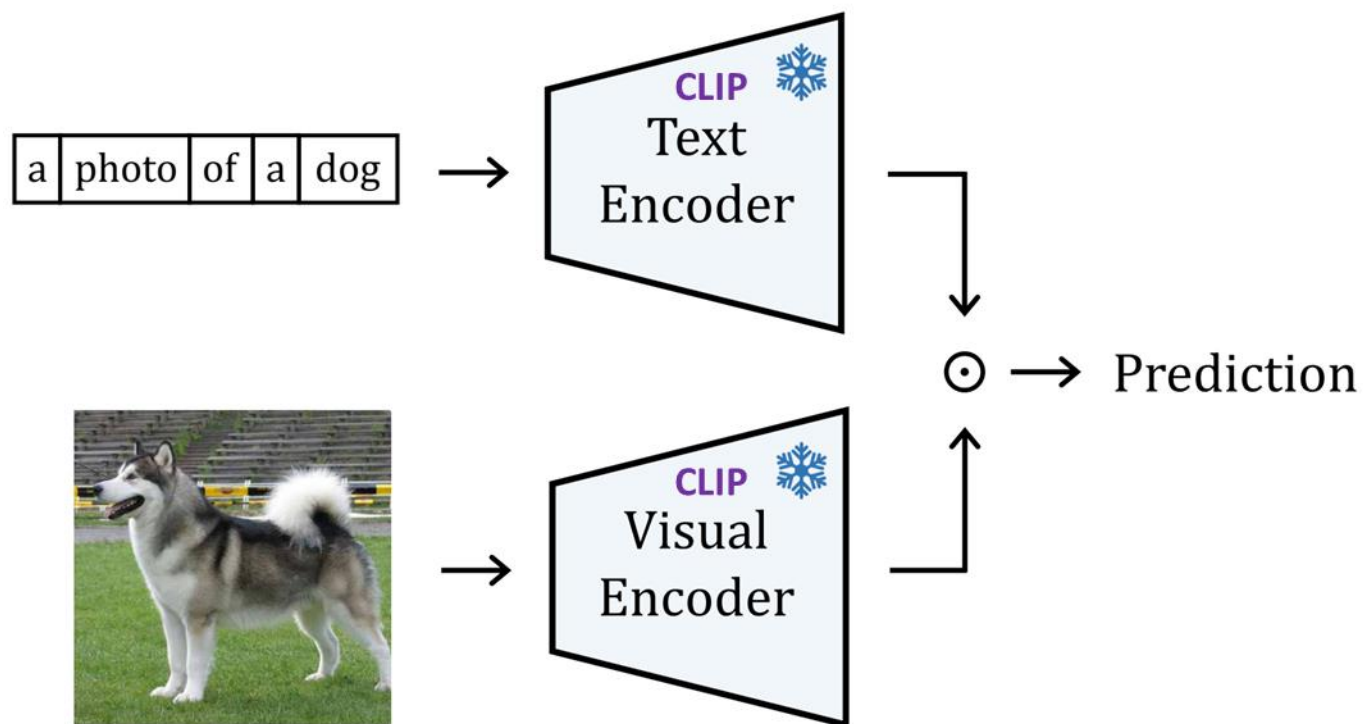
Stage 1: Baselines

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Stage 1: Baselines

- Few-Shot Classification
- MNIST, CIFAR, Mini-Imagenet
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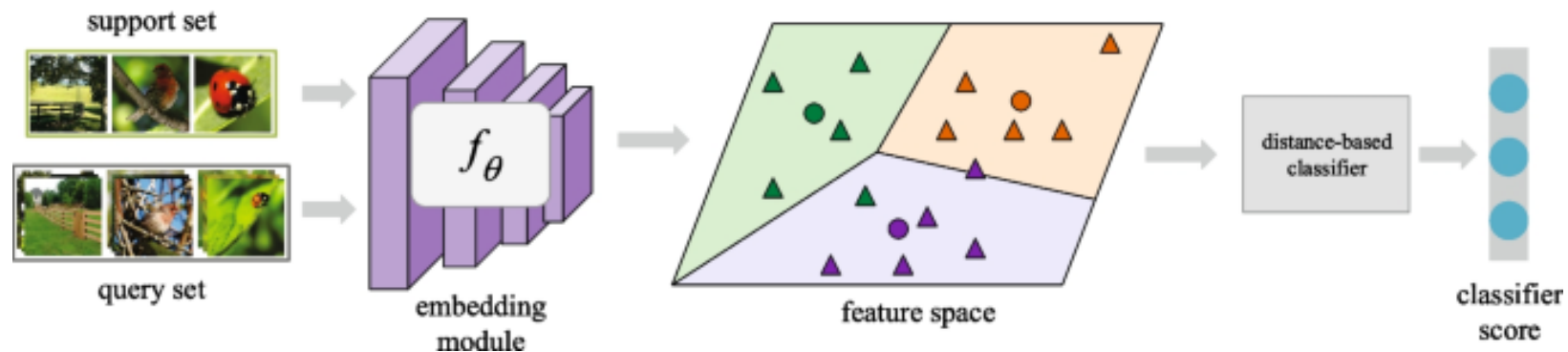


Stage 2: FM as a last layer

- K-means Prototype based
- Few-shot CLIP
- standard FM training
- vs. rolled-out training

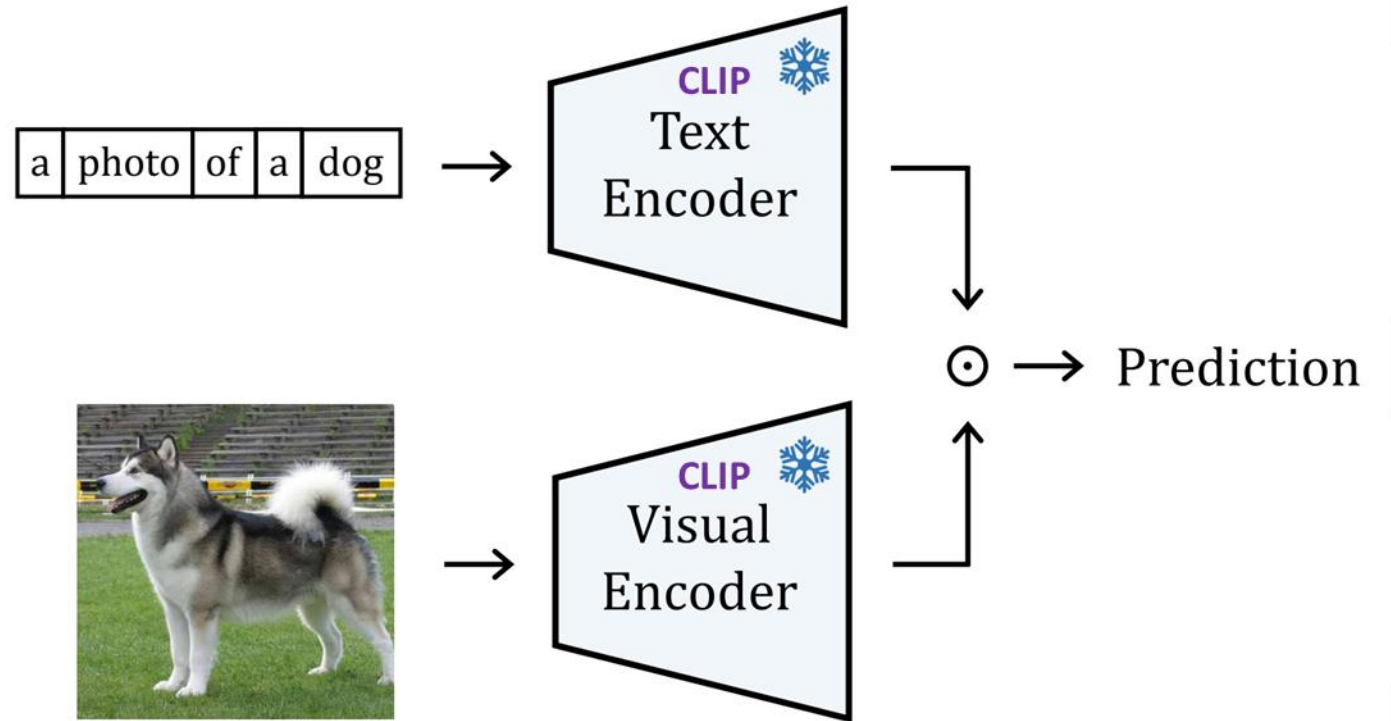
Stage 2: FM as a last layer

- **K-means Prototype based**
- Few-shot CLIP
- standard FM training
- vs. rolled-out training



Stage 2: FM as a last layer

- K-means Prototype based
- **Few-shot CLIP**
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Stage 2: FM as a last layer

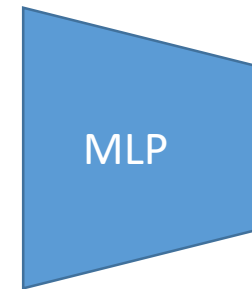
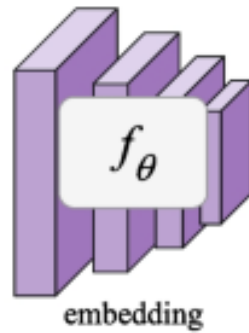
- K-means Prototype based
- Few-shot CLIP
- **standard FM training**
- vs. rolled-out training

Stage 2: FM as a last layer

- K-means Prototype based
- Few-shot CLIP
- standard FM training
- **vs. rolled-out training**

Stage 3: FM prior to last layer

- before Linear Classifier
- revised training scheme



Stage 4: Extensions

- Structured tasks (e.g. segmentation)
- Encoder fine-tuning

